Wavelet Transformation-Based Fuzzy Reflex Control for Prosthetic Hands to Prevent Slip

Hua Deng, *Member, IEEE*, Yi Zhang, and Xiao-Gang Duan, *Member, IEEE*

***Abstract*—A wavelet transformation-based fuzzy reflex control method is proposed to prevent slip and maintain stable grasping posture for prosthetic hands. A key component of the control method is that the response time from slip detection to slip prevention is very short. The proposed control method mainly includes slip detection, reflex grasping force estimation and use of a reflex controller. In the slip detection, initial slip information can be rapidly obtained via wavelet transformation as soon as disturbances cause the grasped object to slip. This shortens the response time of the reflex control. The initial slip information can then trigger the reflex control. In the grasping estimation, a reflex force estimation model is presented to estimate a force increment that is used to adjust to the desired force. In the reflex controller, a fuzzy logic controller is adopted to track the desired force. Finally, the proposed control method is applied to control a prosthetic hand with a single degree of freedom. The experimental results show that the proposed control method can adjust the grasping force and effectively prevent slip under disturbances. The time of slip prevention is less than 160 ms.**

***Index Terms*—Fuzzy logic control, Reflex control, prostheses, slip prevention, wavelet transformation.**

I. Introduction

An important requirement is that the grasping force control must be reliable in the design of prosthetic hands. Amputees hope that prosthetic hands can prevent slip and stably grasp objects under disturbances, such as filling water in a grasped cup, drinking water, etc. [15, 21]. A large number of studies show that proprioceptive reflexes can regulate movement and posture [1-2]. A person with abnormal muscle tone, including spasticity, dystonia or Parkinson's disease, cannot fully control their movements [3-4]. When a healthy person grasps an object, he can continuously adjust the applied forces to the object based on the anticipated force and the reflex control, which depends on tactile feedback.

Recently, various biological signal decoding methods have been applied to detect human movement intention using electromyographic (EMG) or electroencephalogram (EEG) signals [5-6]. These decoded signals are typically used to control prostheses. Although the biological signal decoding can control prosthetic hands, the following challenges still currently remain: (1) Accurate grasping force of a finger cannot be achieved, although many decoding methods have been presented in the past [7-12] in which grasping patterns have often been achieved. Inaccurate grasping force may cause instability in prosthetic hands grasping objects. (2) Biological signal decoding invariably introduces a time delay between the time when the amputee desires to control the prosthetic hand and the time when the prosthetic hand reacts to the amputee. (3) Prosthetic hands have no proprioception feedback to amputees, though there have been notable research efforts, such as studies on haptic [13] and vibrotactile feedback [14]. However, even if these feedbacks can stimulate amputees to adjust to grasping force, the time delay would be longer than that of a natural hand [15]. Thus, to compensate for the influence of outside forces and the delay time that comes from biological signal decoding methods, a good option is to develop a low-level reflex control strategy.

A key component of slip prevention is the ability to detect slippage, adjust force and control the object. As a grasped object slides from prosthetic hands, slip detection and prevention become a challenge [15-18]. Slip detection methods [19] mainly include the ratio of vertical strain/stress to the tangential strain/stress [18, 20], the ratio of normal to tangential forces [21], the elastic deformation of the slip of a grasped object [22-24], the derivative of the grasping force [17, 25], pressure conductive rubber [26], etc. Among these methods, the force sensitive resistor (FSR) sensor is low-cost and can be used to measure the grasping force [17, 26, 48]. Furthermore, initial slip can be extracted from the grasping force. Therefore, the FSR sensor is adopted in this paper. Many researchers have made outstanding contributions to prosthetic hand control [27-31], such as how to adjust the grasping force and prevent slip. They mainly include three control methods: proportional derivative (PD) slip prevention control [15], proportional positional control mode [21] and the model-based control method [15, 24, 31]. The scheme of PD slip prevention control is used with Otto Bock's SensorHand in which the grasped object slip is not actually detected [15]. Rather, the applied grasping force is increased proportionally to the measured shear forces. In the proportional positional control mode, the

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time of slip prevention is much slower than that of biological latencies observed in humans in adjusting grasp. In the model-based control, the time of slip prevention may be slower than that of human reflexes because model-based control is relatively complex. Moreover, those control methods require additional tangential forces [15, 21] or positions [24, 31]. The control of prosthetic hands faces certain challenges in nonlinearities caused by the existence of motor dead bands, friction and large gear ratios [32]. Most of all, unknown disturbances may cause the grasped object to slide. Even more serious is the fact that the grasped object may drop from prosthetic hands. The aforementioned control methods of slip prevention have their disadvantages, such as the long delay time [21], the fact that the grasped object under investigation is not a daily supply but an experimental device [15], the long setting time of fine-tuning the gripping force of the robot hand [26], etc.

The concept of fuzzy logic control (FLC) was originally introduced [33] and applied in an attempt [34] to control systems that are structurally difficult to model. Since then, FLC has been an extremely active and fruitful research area with many industrial applications reported [35-39]. Humans can adjust grasping force in reflex control based on slip speed or strength [40]. The grasping force of an individual may be different from that of others when grasping the same object. In other words, the grasping force of a natural hand is uncertain. The fuzzy logic controller has two advantages. First, an accurate mathematical model of the controlled system is not required, and second, a satisfactory nonlinear controller can often be developed empirically in practice without complicated mathematics [41]. Thus, fuzzy logic control may be a good option for slip prevention as it imitates human reflex control. Although some researchers have applied the fuzzy logic controller to prosthetic hands for general grasping [42-43], fuzzy logic controllers have not been applied for reflexes.

Motivated by previous discussion, a wavelet transformation-based fuzzy reflex control method is proposed to avoid slippage for prosthetic hands. Based on initial slippage caused by disturbances, a slip detection and reflex force estimation model are presented to respectively catch slippage information and estimate grasping force. In the slip detection, slippage information is extracted from the grasping force by wavelet transformation. In the reflex force estimation model, a fuzzy logic system is designed to estimate and adjust the reflex grasping force. Based on the estimated force, a fuzzy logical controller is adopted to track the desired grasping force. Finally, the wavelet transformation-based fuzzy reflex control method is applied to control a prosthetic hand with a single degree of freedom. The experimental results show that the proposed control method can effectively adjust grasping force and prevent slip under disturbances.

II. PROBLEM FORMULATION

The dynamics of a prosthetic hand with a single degree of freedom can be described by [32]

$$M\ddot{\theta}+C\dot{\theta}+D\theta=Bu-lf \tag{1}$$

where, θ, $\dot{\theta}$ and $\ddot{\theta}$ denote the joint angle, angular velocity and angular acceleration, respectively; M is the equivalent inertia; C represents the effective damping of the motor and linkage system in addition to the back electromotive force terms; D is the system's stiffness; B is the quotient of the torque constant and armature resistance of the motor; u is the voltage input to the motor of the prosthetic hand; f is the force exerted by the environment; and l is the distance between the contact point and rotational joint.

In general, two methods exist for the control of a prosthetic hand: biological signal decoding and force feedback control. The former is an open loop control, as shown in Fig. 1, where human motion information is decoded and used to control prosthetic hands through neural, EMG or EEG signals [5-9]. However, the accurate grasping force of humans cannot be obtained because of low-level decoding technology. Moreover, amputees focus on the grasping process all the time due to the lack of force feedback, resulting in fatigue. A person can prevent slip and stably grasp an object under disturbances through reflex control. Thus, another method is reflex control. It can reduce unnecessary energy consumption and the chances of damaging the object or poorly directing the grasping forces [21]. The biological latency of reflex control is very short. The reflex control approximates the 70-ms latencies observed in humans in adjusting grasping forces [53]. However, the time of slip prevention is much slower than that of biological latencies observed in humans, although some slip prevention does occur [15, 21, 24, 31]. In the literature [21], the delay time is approximately 750 ms.

Motivated by the previous discussion, a wavelet transformation-based fuzzy reflex control method is attempted to adjust grasping forces and prevent slip with minimal time delay.

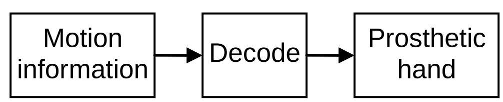

Fig. 1. Schematic structure of prosthetic hand control.

III. WAVELET TRANSFORMATION-BASED FUZZY REFLEX CONTROL

The biological reflex control [2, 44] is as shown in Fig. 2, where tactile senses obtain slip information, and interneurons in the spinal cord estimate proper grasping force. Motor neurons can control the joint dynamics through controlling muscle dynamics.

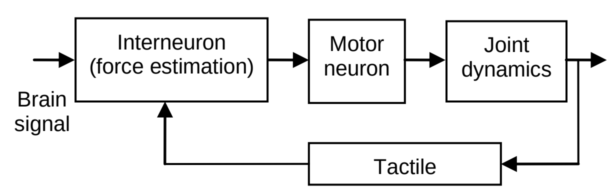

Fig. 2. Schematic structure of the biological reflex control.

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Supposing that the impedance between the prosthetic hand and the environment is an elastic element. The contact force *f* in (1) is expressed by

$$f = K_{ev} x \tag{2}$$

where K_{ev} is environment stiffness and $x = \tilde{x} - x_0$, where $\tilde{x}$ denotes the real value, and x_0 denotes the contact position when the prosthetic hand is initially in contact with the environment. Thus, the displacement x can be approximated by

$$x = l\theta \tag{3}$$

where l is the distance between the contact point and the rotational joint.

For simplicity, we consider that force is applied to only one direction. Based on impedance control [45], integrating (2) and (3), equation (1) becomes

$$\ddot{f} = a_1 \dot{f} + a_0 f + c\tau \tag{4}$$

where $a_1 = -C/M$, $a_0 = -(D + l^2 K_{ev})/M$ and $c = BlK_{ev}/M$.

By using an FSR, a reflex grasping force estimation model and a fuzzy logic-based reflex controller are used to replace the tactile senses, interneurons and motor neurons, and the fuzzy logic-based reflex control is proposed as shown in Fig. 3, where f_d^*, f_d and f denote the initial, desired and real grasp forces, respectively, $e = f_d - f$ is the error, q denotes the slippage information, and f_e is the estimated force. The control method mainly includes three units, i.e., slip detection, reflex grasping force estimation and the reflex controller.

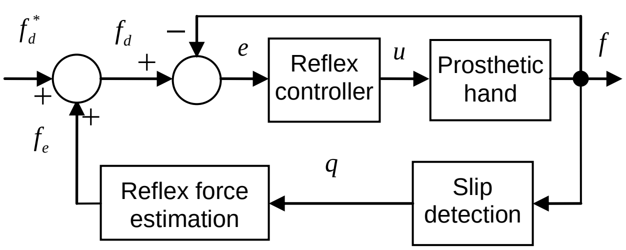

Fig. 3. The fuzzy logic-based reflex control scheme of the prosthetic hand.

A. Initial Slip Detection

To detect the initial slip, the FSR placed at the tip of the finger is used to detect the grasping force. According to the theory of friction vibration, friction force can be vibrated at low speed [46-47]. The vibration of friction force causes the grasping force to vibrate. The slippage information can be extracted from the grasping force through wavelet transformation [48, 55] when the object initially slips. Grasping force *f* can be expressed by

$$f(t) = \sum_{n=t_0}^{t_0+N} c_n \varphi(2^{j_0} t - n) + \sum_{n=t_0}^{t_0+N} \sum_{j=j_0}^{J} d_{j,n} \psi_{j,n}(2^j t - n) \tag{5}$$

where $\varphi(t)$ and $\psi(t)$ are the scaling function and wavelet, respectively; c_n and $d_{j,n}$ are the real-valued expansion coefficients, n is the time translation, j is the scale, and N, J, t_0 and j_0 are positive integers.

$d_{j,n}^2$ denotes wavelet coefficient energy [53]. Let

$$q = \begin{cases} 1 & d_{j,n}^2 \geq c_s \\ 0 & d_{j,n}^2 < c_s \end{cases} \tag{6}$$

where c_s is a positive threshold. If a grasped object slips, then $q = 1$, or else $q = 0$.

The threshold c_s can be selected empirically. Let

$$c_s \geq N_0 c_0 \tag{7}$$

where c_0 denotes the coefficient of static grasping force, and $N_0 \geq 5$.

Example: a prosthetic hand grasps a metal cup. After stable grasping is achieved, a disturbance is applied on the cup, causing sliding. The initial slip is caught by wavelet transform as soon as the object initially slips, as shown in Fig. 4, where the threshold $c_s = 0.003$, the sampling time is 1 ms, and $N = 16$. This suggests that the initial slip information can be obtained in 16 ms after the object slips.

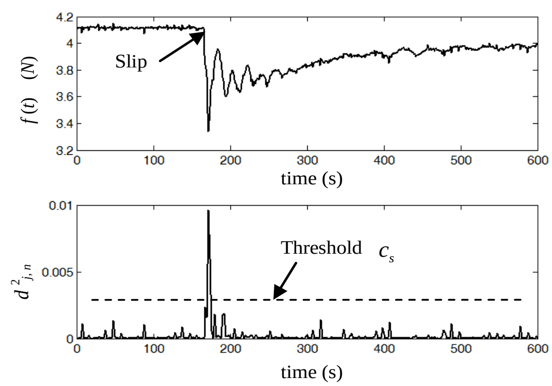

Fig. 4. Slip detection based on wavelet transformation.

B. Reflex Grasping Force Estimation

If the object slips, this indicates that the initial grasping force is not proper and should be adjusted. Considering that the natural human grasping force in reflex control is vague, a fuzzy logic system is proposed to estimate the reflex grasping force when the object slips from a prosthetic hand.

Based on equation (5), let

$$y(\tau) = \frac{1}{N} \sum_{n=0}^{N} \left| d_{j,n} \right|$$
$$\Delta y(\tau) = \frac{y(\tau) - y(\tau - 1)}{\Delta T} \tag{8}$$

where $d_{j,n}$ is given in (5), τ denotes time, and ΔT is the sampling period.

There are no system methods to design the fuzzy logic system. Here, the main focus is the basic principle of the

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grasping force estimation. For simplicity, a fuzzy logic system for grasping force estimation is presented, as shown in Fig. 5, where $E = c_l y$, $R = c_h \Delta y$, and c_l and c_h denote coefficients of input variables y and Δy, respectively. The standard triangular input and output membership function (MF) are designed, as shown in Fig. 6 (a) and (b), respectively, where L and H are the half membership degree domain of input MF and output MF, respectively. The fuzzy rule is adopted:

$$\text{If } E \text{ is } A_i \text{ and } R \text{ is } B_j \text{, then } \tilde{f} \text{ is } G_{i+j}$$

where E and R, A_i and B_j ($i, j = -1, 0, 1$) denote input variables and input fuzzy sets, respectively; $\tilde{f}$ and G_{i+j} denote the estimating force and output fuzzy set, respectively. The details of the rules are shown in Table I, where the linguistic labels are negative small (NS), zero (ZO), positive small (PS), and positive medium (PM).

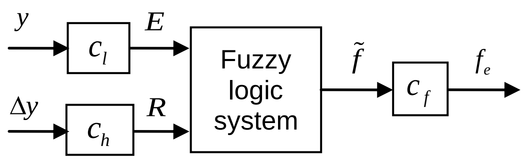
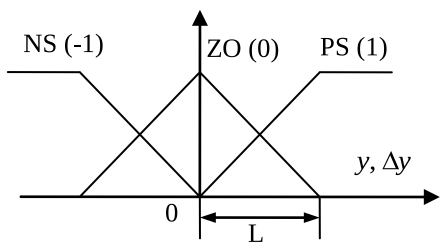
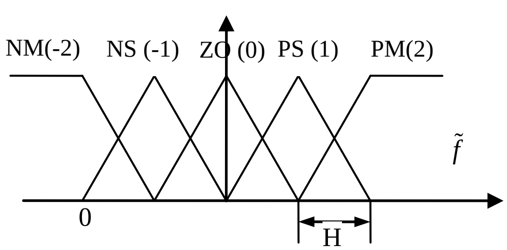


Fig. 5. Fuzzy logic system for grasping force estimation.



(a) Input membership function



(b) Output membership function

Fig. 6. Membership functions.

The inference adopts a standard Mandani inference engine. The center of sets method is used to obtain a crisp output [51]. The crisp output is

$$\tilde{f} = \frac{\sum_{m=1}^{M} g^m \left[\mu_{A_i^m}(y) * \mu_{B_j^m}(\Delta y) \right]}{\sum_{m=1}^{M} \left[\mu_{A_i^m}(y) * \mu_{B_j^m}(\Delta y) \right]} \tag{9}$$

where g^m is the centroid of the consequent set of the mth fired rule, M denotes the number of fired rules, $\mu_{A_i^m}(y)$ and $\mu_{B_j^m}(\Delta y)$ denote the membership functions of input y and Δy, respectively, and '*' denotes the t-norm operator.

By integrating (6), the output of the estimation of reflex grasping force is

$$f_e = \begin{cases} c_f \tilde{f} & q = 1 \\ 0 & q = 0 \end{cases} \tag{10}$$

where c_f is a positive constant.

Equation (10) is actually an increment of the grasping force. Thus, the desired grasping force is amended to

$$f_d = f_d^* + f_e \tag{11}$$

TABLE I
FUZZY RULES

R/E	A_{-1} (NS)	A_0 (ZO)	A_1 (PS)
B_1 (PS)	G_0 (ZO)	G_1 (PS)	G_2 (PM)
B_0 (ZO)	G_{-1} (NS)	G_0 (ZO)	G_1 (PS)
B_{-1} (NS)	G_{-2} (NM)	G_{-1} (NS)	G_0 (ZO)

C. Reflex Controller

A prosthetic hand has nonlinearities caused by the existence of motor dead bands, friction and large gear ratios [32]. Moreover, the grasped object is unknown. This uncertainty can be described by fuzzy systems [49-50]. FLC has two advantages: 1) it has robustness due to its inherent saturation; and 2) it does not depend on an accurate model of the controlled object. Thus, FLC is chosen as the reflex controller. The detailed design of the FLC has been discussed in previous studies [41, 51]. Here, only the structure of the FLC is as shown in Fig. 7. One two-dimensional rule base is used, which is normally chosen as a linear rule base.

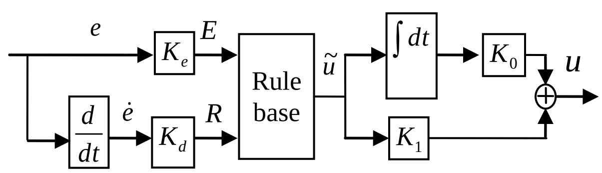


Fig. 7. Structure of the fuzzy controller with $K_d = \alpha K_e$, $K_1 = \beta K_0$, and $e = f_d - f$.

The model of the FLC is given as [41, 51]

$$u = u_e + K_1 sat(\sigma) \tag{12}$$

with

$$u_e = K_0 \int \tilde{u} dt = \alpha K_0 K_e \left(e + \frac{1}{\alpha} \int e dt \right) + \Delta u$$

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where $\sigma = E + R = K_e e + K_d \dot{e}$; $\tilde{u} = sat(\cdot)$ is a saturation function, Δu is an error; K_e , K_d , K_0 , K_1 , α and β are positive constants.

The design parameters of the FLC are given in Appendix.

IV. EXPERIMENTS

The proposed fuzzy logic-based reflex control method was applied to control a prosthetic hand with a single degree of freedom, as shown in Fig. 8. The prosthetic hand is used to grip different objects. The motor of the prosthetic hand adopts a DC-Micromotor (series 2657). The detail of the prosthetic hand is given in reference [54]. In this process, a key task is to adjust the grasping force and prevent slip in the event that an environmental perturbation occurs. An FSR (Interlink Electronics) mounted on the surface of the thumb is used to measure the grasping force. Its characteristics are given as follows: hysteresis (+10%), device rise time (3 ms) and force resolution (continuous). Further details can be obtained from Interlink Electronics. The basic control scheme is shown in Fig. 9. The control algorithm is developed in NI Labview. The total computational expense is less than 8 ms.

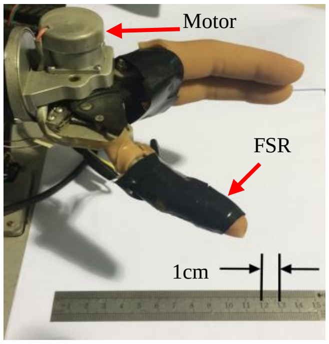

Fig. 8. A single DOF prosthetic hand.

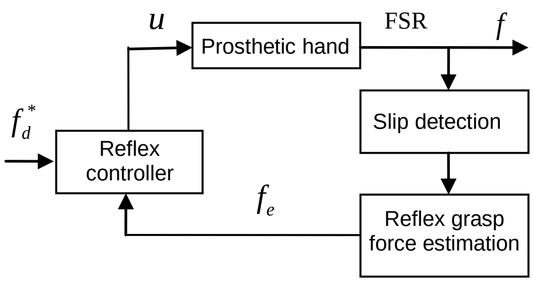

Fig. 9. Control diagram of the prosthetic hand.

To imitate disturbances such as filling water or other things into a grasped cup, a disturbance generator is used, as shown in Fig. 10. When a grasped object is placed under the striker, the bar is pulled upward, and the striker produces an impulse force on the grasped object. The impulse force can be obtained from the force meter. The scope of the impulse force is 0-5 Newtons.

The mother wavelet must be orthogonal and symmetric. Thus, wavelets such as haar, Daubechies (db1), sysmlets (sym1), etc., can be used. The order is a one-dimensional wavelet. In this experiment, the one-dimensional haar wavelet is adopted. The sampling period is 1 ms for slip detection. The total control period is 10 ms. The parameters in (5) are set as $J = 1$ and $N = 16$. The estimation model parameters of the reflex grasping force are set as $c_l = 1$, $c_h = 0.2$, $c_f = 1.5$, $L = 0.1$ and $H = 1/3$, based on experiments of grasping paper cups, smooth plastic bottles and metal cups. The reflex control parameters are set as $K_e = 1$, $K_d = 0.02$, $K_0 = 0.01$ and $K_1 = 1.2$.

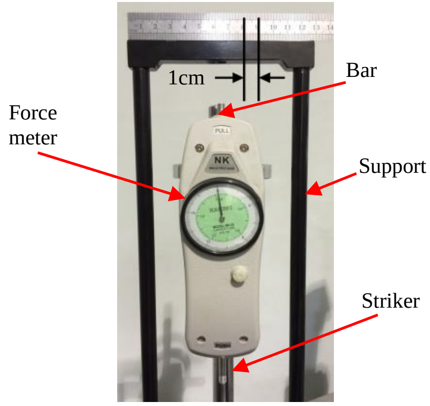

Fig. 10. Disturbance generator.

Considering that PID slip prevention control is used with Otto Bock's SensorHand [15], a PID controller is selected to replace the fuzzy logic controller in (11). Based on the relation of the FLC and PID controller [41], the PID controller parameters are set as $\tilde{K}_p = 1.2$, $\tilde{K}_i = 0.01$ and $\tilde{K}_d = 0.24$, where $\tilde{K}_p$, $\tilde{K}_i$ and $\tilde{K}_d$ denote the proportional, integral and derivative gain, respectively.

The grasped object includes three objects: paper cups, smooth plastic bottles and metal cups, as shown in Fig. 11. The experimental procedures are described as follows:

1) The prosthetic hand stably grasps an object at initial grasping force f_d^*.
2) The disturbance generator produces a certain disturbance that is exerted on the top of the grasped object. The grasped object slips.
3) After slip is detected, the grasping force estimation model produces an increment f_e. Thus, the desired grasping force is revised to $f_d = f_d^* + f_e$.
4) Based on the updated grasping force, the prosthetic hand is controlled by the reflex controller.

The control performances are shown in Fig. 12 to Fig. 14. These figures show that the proposed slip prevention method can detect the initial slip, adjust grasping force and avoid grasped object dropping. If there is no slip prevention control, the grasped object was dropped, as shown in Fig. 15. These experimental results show that the proposed slip prevention can effectively avoid object dropping while disturbance occurs. The experimental results show that the control performance of the fuzzy logic controller is better than that of the PID controller.

The grasping posture, contact position, grasping force, friction coefficient, etc., may cause converged speed of the grasping force to the desired force. Those influences are a

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complex and nonlinear process. It is very difficult to analyze it by using a dynamic model. For the smooth plastic bottle, the rise time is relatively slower, as shown in Fig. 13. One possible reason is that the friction coefficient and grasping force are relatively smaller than that of the metal cup.

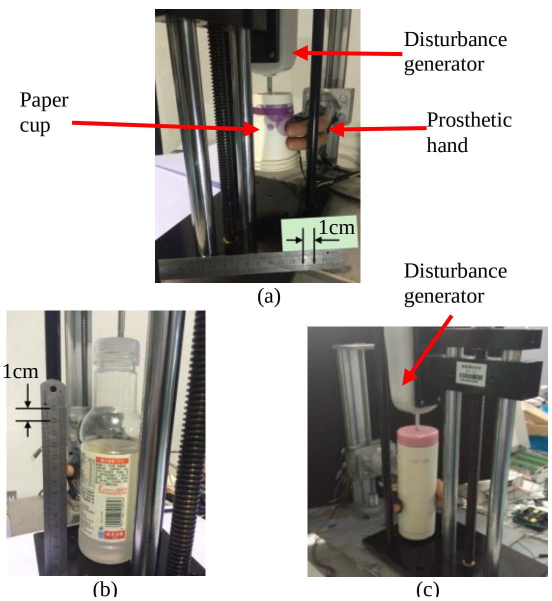

Fig. 11. The prosthetic hand grasps: (a) a paper cup, (b) a plastic bottle, (c) a metal cup.

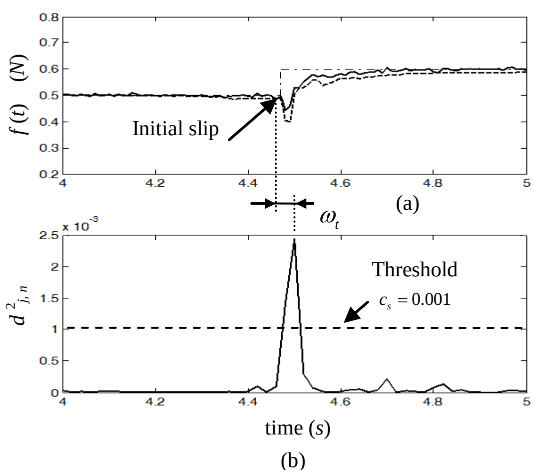

Fig. 12. Performance of slip prevention for the paper cup: (a) grasping force (solid: proposed, dotted: PID, dash-dot: desired); (b) coefficient.

The slip detection time ω_t is defined, as shown in Fig. 12. This is the time from the initial slip to the maximum value of coefficient $d^2_{j,n}$. In other words, the FSR can detect slip information after the grasped object initially slips. The experimental results are shown in Table II, where ϖ_t is the average of the slip detection time, σ_t is the variance, and p_i is the perturbation, where $p_1 = 1N$, $p_2 = 2N$, and $p_3 = 4N$. The experiment for each cup is repeated eight times for each perturbation p_i ($i = 1,2,3$). The statistical results are shown in Table II, where the slip detection time is less than 10 ms. This is the reason that the sampling period is 1 ms.

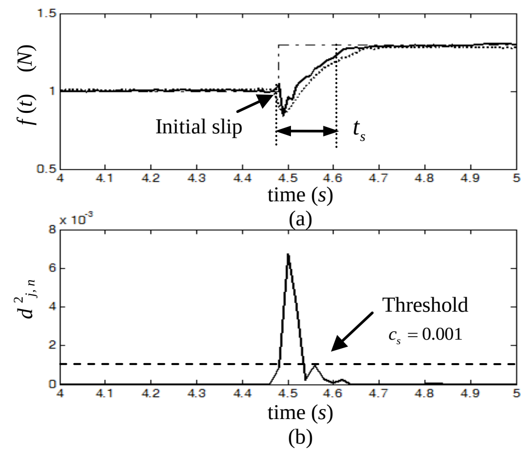

Fig. 13. Performance of slip prevention for the plastic bottle (a) grasping force (solid: proposed, dotted: PID, dash-dot: desired); (b) coefficient.

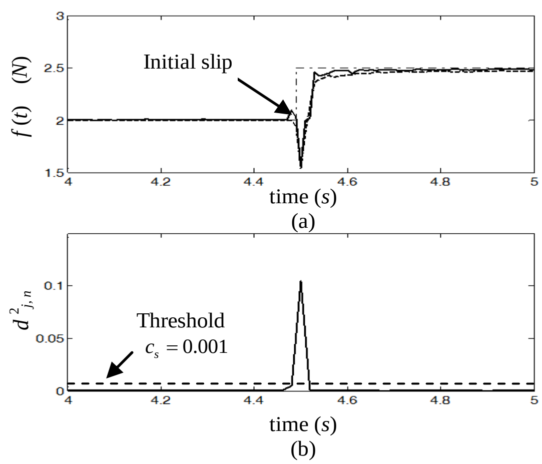

Fig. 14. Performance of slip prevention for the metal cup: (a) grasping force (solid: proposed, dotted: PID, dash-dot: desired); (b) coefficient.

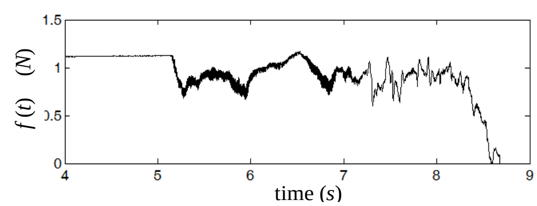

Fig. 15. The cup was dropped without slip prevention control.

TABLE II
THE TIME OF SLIP DETECTION

Object	Paper cup			Plastic bottle			Metal cup		
p_i	p_1	p_2	p_3	p_1	p_2	p_3	p_1	p_2	p_3
ϖ_t (ms)	7.3	6.0	4.7	6	5.3	4.7	6.3	5	4
σ_t (ms)	0.2	0.7	0.6	0.7	0.8	0.8	0.9	0.7	0.7

The slip prevention time t_s is defined as shown in Fig 13. This is the time from the initial slip to the end of the slip. Here, the object stops slipping when the grasping force reaches 90% of the desired force. The control experiment for each object is

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performed eight times for each perturbation p_i $(i=1,2,3)$. The average time $\bar{t}_s$ and variance ς are given in Table III, where the time of slip prevention is less than 160 ms. This approximates to the 70-ms biological latencies observed in humans in adjusting grasp [52], which is far less than 750 ms in reference [21]. If the proposed control algorithm is used in an embedded system, the slip prevention time will be further shortened.

TABLE III
THE TIME OF SLIP PREVENTION

Object	Paper cup			Plastic bottle			Metal cup		
p_i	p_1	p_2	p_3	p_1	p_2	p_3	p_1	p_2	p_3
$\bar{t}_s$ (ms)	57	83	67	60	143	143	60	97	100
ς (ms)	4.7	4.7	4.7	1	9	17	14	12	16

Remarks 1: The proposed reflex control can prevent an object from slipping as long as the initial slip is detected. Slip detection is similar to a switch that triggers the reflex control. Thus, the slip detection is very important. The initial slip information can be obtained as an object slips based on reference [46], where different directions are considered. Moreover, the mechanism of slip detection is investigated in our recent published paper [55], where the initial slip information can be obtained as soon as the object slips based on the frictional vibration [46-47]. These works imply that the proposed reflex control can stably grasp objects under different perturbations, such as liquid moving inside the cup, lifting objects, etc.

Remarks 2: The biomimetic tactile sensing method used for slip prevention [21] has a long delay time. The delay time is 750 ms. In [15], the grasped object used for grasping force control is not a daily supply, but rather an experimental device. In [26], the settling time of fine-tuning the gripping force of the robot hand is also long. The proposed reflex control method can prevent object slippage in less than 160 ms. The time of slip prevention should be further reduced if the reflex control is realized in a digital signal processor (DSP).

V. CONCLUSION

In this paper, the wavelet transformation-based fuzzy reflex control method was designed to adjust grasping force and avoid slippage for prosthetic hands under disturbances. A slip detection and reflex force estimation model were presented to obtain slippage information and estimate grasping force, respectively. Based on the estimated force, a fuzzy logical controller was adopted to track the desired grasping force. Finally, the proposed control method was applied to control a prosthetic hand with a single degree of freedom to grasp different objects. The experimental results show that slip detection can detect initial slippage. The reflex force estimation model can estimate a proper force that is used to adjust the grasping force. The wavelet transformation-based fuzzy reflex control can effectively prevent slippage under disturbances. The proposed method is compared with the PID controller. Experimental results show that the control performance of the proposed control method is better than that of the PID controller, and the time of slip prevention is less than 160 ms.

APPENDIX

Lyapunov's stability theory is used to construct algorithms for parameter design. The Lyapunov function is shown as follows:

$$V(\sigma)=\frac{1}{2K_d}\sigma^2 \tag{13}$$

The derivative of $V(\sigma)$ is

$$\dot{V}(\sigma)=\frac{1}{K_d}\sigma\dot{\sigma} \tag{14}$$

where

$$\dot{\sigma}=K_d(\frac{1}{\alpha}\dot{e}+\ddot{e}) \tag{15}$$

Substituting $e=f_d-f$, $f_d=f_d^*+\Delta f_d$ and (12) into (4), yields

$$\ddot{e}=\left[a_1\dot{e}+a_0e-c\alpha K_0K_1\left(e+\frac{1}{\alpha}\int edt\right)\right]-\tilde{f}_d-\tilde{f}_e-c\Delta u-K_1sat(\sigma) \tag{16}$$

where $\tilde{f}_d=a_1\dot{f}_d+a_0f_d-\ddot{f}_d$, $\tilde{f}_e=a_1\dot{f}_e+a_0f_e-\ddot{f}_e$.

Assume $\|\tilde{f}_d\|\le F_1$ and $\|\tilde{f}_e\|\le F_2$, where F_1 and F_2 are two finite positive numbers. Assume $\|\Delta u\|\le u_0$, where u_0 is a finite positive number. Using linear control theories [48], parameters K_e , K_0 and α can be designed so that the nominal linear system in the square brackets of (16) is stable.

By substituting (16) into (15), one can obtain

$$\begin{aligned}\dot{\sigma}=K_d\Bigg[&\left(\frac{1}{\alpha}\dot{e}+a_1\dot{e}+a_0e-c\alpha K_0K_e(e+\frac{1}{\alpha}\int edt)\right)\\&-\tilde{f}_d-\tilde{f}_e-c\Delta u-cK_1sat(\sigma)\Bigg]\end{aligned} \tag{17}$$

Suppose that $c\alpha K_0K_e(e+(1/\alpha)\int edt)$ can compensate most of the undesirable effects. Thus, one has

$$\left\|\left(\frac{1}{\alpha}\dot{e}+a_1\dot{e}+a_0e-c\alpha K_0K_e(e+\frac{1}{\alpha}\int edt)\right)-\tilde{f}_d-\tilde{f}_e-c\Delta u\right\|\le\delta \tag{18}$$

where δ is a positive number.

In the saturation region, i.e., $|\sigma|>1$, by integrating (18), (17) becomes

$$\dot{\sigma}=K_d\left(\delta-cK_1\operatorname{sgn}(\sigma)\right) \tag{19}$$

Then

$$\dot{V}(\sigma) = \frac{1}{K_d}\sigma\dot{\sigma} \le \delta\sigma - cK_1 |\sigma| \tag{20}$$

When $\sigma < -1$, the stability requirement of $\dot{V}(\sigma) < 0$ is guaranteed. When $\sigma > 1$, one should have $cK_1 |\sigma| > \delta\sigma$ to meet the stability requirement of $\dot{V}(\sigma) < 0$, which yields

$$K_1 > \frac{\delta\sigma}{K_1 |\sigma|} = \frac{\delta}{K_1} \tag{21}$$

Thus, whenever $|\sigma(0)| > 1$, $\sigma(t)$ will strictly decrease until it reaches the non-saturation region of $\{|\sigma| \le 1\}$ in finite time and remains inside thereafter. In the non-saturation region, based on $\sigma = K_e e + K_d \dot{e}$, one has

$$\dot{e} = \frac{K_e}{K_d} e + \frac{1}{K_d}\sigma \tag{22}$$

Select the Lyapunov function $V(e) = K_d e^2 / 2$. Its derivative is

$$\dot{V}(e) = -K_e e^2 + e\sigma \le -K_e e^2 + |e| \le -(1-\lambda) K_e e^2 \tag{23}$$

where $0 < \lambda < 1$. Thus, in the non-saturation region of $|\sigma| \le 1$, the trajectory reaches the set $\{|e| \le 1/(K_e \lambda)\}$ in finite time, which implies that the closed-loop control system is stable.

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